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Rotation-Vibration Spectra of Molecules

FTIR Calibration, Interferogram Processing, Cuvette Interference, and HCl Spectral Analysis

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Abstract

This report presents a Fourier-transform infrared spectroscopy investigation of molecular rotation-vibration spectra using the PerkinElmer Spectrum 100 FTIR spectrometer. The wavenumber calibration was first checked using water-vapour and polystyrene absorption bands. The mean deviations were found to be $\Delta\tilde{\nu}_{\text{water}} = 0.49/\text{cm}$ and $\Delta\tilde{\nu}_{\text{polystyrene}} = 0.78/\text{cm}$, indicating a sufficiently accurate wavenumber scale for the subsequent analysis. Water-vapour interferograms recorded over different mirror ranges showed that longer interferograms provide improved spectral resolution, while zerofilling mainly increases spectral point density and apodisation reduces leakage artifacts at the cost of line broadening. The blocking-filter measurements gave residual signals of approximately 0.05 % for glass and 0.72 % for NaCl in their selected blocking ranges. From the interference fringes of an empty NaCl cuvette, gap widths of (47.59 ± 0.25) µm and (46.91 ± 0.19) µm were obtained for the 1/cm and 4/cm spectra, respectively. The background-corrected HCl spectrum showed the expected P- and R-branch structure with isotope splitting due to $H^{35}Cl$ and $H^{37}Cl$. The pure vibrational transitions were determined as $2888.50/\text{cm}$ and $2886.44/\text{cm}$, with rotational constants of $10.31/\text{cm}$ and $10.28/\text{cm}$. The corresponding force constants were approximately $482\,\text{N/m}$ and the bond lengths were about 1.29×10^{-10} m. The isotope abundance analysis gave 71.5 % for ^{35}Cl and 28.5 % for ^{37}Cl. Finally, P-branch intensity modelling yielded effective temperatures of (363.1 ± 35.1) K and (445.5 ± 61.8) K, which reproduce the qualitative intensity trend but are limited by baseline correction, peak overlap, finite resolution, and possible saturation effects.

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1. Introduction

Infrared spectroscopy is a widely used experimental method for investigating the interaction between electromagnetic radiation and matter. In this technique, infrared light is guided through a sample and the transmitted intensity is measured as a function of wavelength or, more commonly, wavenumber. Absorption minima in the transmission spectrum correspond to transitions between molecular or material energy states, while the strength of the absorption provides information about the probability of these transitions and the amount of absorbing material. This makes infrared spectroscopy especially useful for studying molecular gases, lattice vibrations, defects, charge carriers, and concentration-dependent absorption processes.

In this experiment, Fourier transform infrared spectroscopy was used to study the rotation-vibration spectrum of hydrogen chloride gas. The measurements were performed with the PerkinElmer Spectrum 100 FTIR spectrometer, which operates in the mid-infrared spectral range. Instead of recording the spectrum directly by scanning one wavelength after another, an FTIR spectrometer first records an interferogram as a function of optical path difference. The corresponding spectrum is then obtained by Fourier transformation. Because of this measurement principle, the quality of the final spectrum depends not only on the spectrometer itself, but also on processing parameters such as interferogram length, zerofilling, apodisation, and phase correction.

Hydrogen chloride is particularly suitable for this experiment because it is a simple diatomic molecule with an infrared-active rotation-vibration spectrum. In the dumbbell model, the molecule can both rotate and vibrate, and these motions lead to discrete quantum-mechanical energy levels. Infrared absorption occurs when the radiation excites allowed transitions between these states. The resulting HCl spectrum consists of two characteristic branches, the P branch and the R branch, which appear on either side of the pure vibrational transition. From the positions of the absorption lines, molecular quantities such as the pure vibrational transition wavenumber, rotational constant, force constant, moment of inertia, bond length, centrifugal distortion constant, and isotope-dependent spectral splitting can be determined.

Before the HCl spectrum can be evaluated reliably, the performance and calibration of the spectrometer must be checked. Therefore, the wavenumber scale was first verified using known water vapour and polystyrene calibration bands. In addition, water vapour interferograms were recorded for different mirror ranges in order to investigate how interferogram length, zerofilling, and apodisation influence the spectral resolution and line shape. The detected signal in the blocking ranges of glass and NaCl filters was also examined. Furthermore, the interference pattern of an empty cuvette was used to determine its gap width by linear regression.

Overall, this experiment combines several central aspects of infrared spectroscopy. It demonstrates how an FTIR spectrum is generated from an interferogram, how instrumental and processing parameters influence spectral resolution, and how molecular constants can be extracted from a measured rotation-vibration spectrum. The HCl measurement therefore provides a compact but physically rich example of how quantum-mechanical molecular structure becomes visible in an experimentally measured infrared spectrum.

2. Theoretical Background

2.1 Infrared Transmission Spectroscopy

Infrared spectroscopy is based on the interaction between infrared radiation and matter. When infrared light passes through a sample, part of the radiation may be absorbed if the photon energy matches an allowed molecular or material transition. In a transmission experiment, the measured quantity is the transmitted intensity $I(\tilde{\nu})$ compared with the incident or reference intensity $I_0(\tilde{\nu})$. The transmittance is therefore defined as

$$T(\tilde{\nu}) = \frac{I(\tilde{\nu})}{I_0(\tilde{\nu})}. \tag{1}$$

Absorption lines appear as minima in the transmission spectrum. Their spectral positions are related to the transition energies, while their depth gives information about the strength of the absorption. In this experiment, this principle is used for the calibration spectra, the blocking-filter measurements, and especially for the rotation-vibration spectrum of HCl gas.

Infrared spectra are usually plotted as a function of wavenumber rather than wavelength. The wavenumber is defined as

$$\tilde{\nu} = \frac{1}{\lambda}, \tag{2}$$

and is commonly given in 1/cm. The relation between wavenumber, frequency, and angular frequency is

$$\tilde{\nu} = \frac{1}{\lambda} = \frac{\nu}{c} = \frac{\omega}{2\pi c}, \tag{3}$$

where c is the speed of light. For wavelength values given in micrometres, the conversion to wavenumber is

$$\tilde{\nu}\,[1/\mathrm{cm}] = \frac{10^4}{\lambda\,[\mu\mathrm{m}]}. \tag{4}$$

2.2 Absorption, Lambert's Law, and Lambert-Beer Law

The decrease of transmitted intensity through an absorbing material is described by Lambert's law,

$$I(\tilde{\nu}) = I_0(\tilde{\nu}) \exp[-\alpha(\tilde{\nu})d], \tag{5}$$

where $\alpha(\tilde{\nu})$ is the absorption coefficient and d is the path length through the material. If the absorbing substance is

diluted or present with a concentration c_{conc}, the relation becomes the Lambert-Beer law,

$$I(\tilde{\nu}) = I_0(\tilde{\nu}) \exp[-\alpha(\tilde{\nu}) c_{\text{conc}} d]. \tag{6}$$

It is often useful to express absorption using the absorbance

$$A(\tilde{\nu}) = -\ln T(\tilde{\nu}) = \ln\left(\frac{I_0(\tilde{\nu})}{I(\tilde{\nu})}\right). \tag{7}$$

In this report, the Lambert-Beer relation is relevant for estimating the abundance ratio of the chlorine isotopes from the HCl absorption lines. If the path length and transition strength are comparable for two isotope lines, the relative absorbance can be used as an estimate of the relative abundance.

2.3 Wavenumber Calibration

Before evaluating unknown spectra, the wavenumber axis of the spectrometer must be checked. This is done using known calibration bands from water vapour and polystyrene. For each calibration band, the deviation between the measured and reference value is calculated as

$$\Delta\tilde{\nu} = \tilde{\nu}_{\text{meas}} - \tilde{\nu}_{\text{ref}}. \tag{8}$$

A small and systematic deviation indicates that the wavenumber scale is reliable. A larger deviation or a clear trend with wavenumber would indicate a calibration problem. This calibration check is important because the later extraction of molecular constants from the HCl spectrum depends directly on accurate peak positions.

2.4 Principle of FTIR Spectroscopy

The measurements were performed using a Fourier transform infrared spectrometer. In an FTIR spectrometer, the light from a broadband infrared source is divided into two beams by an interferometer. The beams travel different optical paths and are recombined, producing interference at the detector. Instead of recording intensity directly as a function of wavenumber, the detector first records an interferogram as a function of optical path difference x.

The optical path difference is defined as

$$x = x_2 - x_1, \tag{9}$$

where x_1 and x_2 are the optical path lengths of the two interferometer arms. For a monochromatic component with wavenumber $\tilde{\nu}$, the interferogram contains an oscillating contribution of the form

$$I(x) = 2S(\tilde{\nu}) \cos(2\pi\tilde{\nu}x), \tag{10}$$

where $S(\tilde{\nu})$ is the spectral intensity. A real infrared source consists of many wavenumber components, so the measured interferogram is a superposition of many such cosine contributions.

2.5 Fourier Transformation of the Interferogram

The spectrum is obtained from the interferogram by Fourier transformation. In the ideal continuous case, the spectrum is given by

$$S(\tilde{\nu}) = \int_{-\infty}^{\infty} I(x) \exp(-2\pi i\tilde{\nu}x)\, dx. \tag{11}$$

In practice, the interferogram is measured only over a finite path range and consists of discrete data points. Therefore, a discrete Fourier transform is used. For N measured points with spacing Δx, the discrete spectral spacing is

$$\Delta\tilde{\nu} = \frac{1}{N\Delta x}. \tag{12}$$

This relation is central for Task 1.2, where spectra are calculated from water-vapour interferograms recorded with different mirror ranges. Increasing the measured path length increases the spectral resolution because narrower spectral features require a longer interferogram to be resolved.

2.6 Spectral Resolution and Interferogram Length

The spectral resolution of an FTIR measurement is determined mainly by the maximum optical path difference. If two spectral lines are very close together, their interferogram contains a slowly varying beating pattern. To resolve the two lines, this beating must be sampled over a sufficiently long path range.

A longer interferogram therefore corresponds to a better spectral resolution. In simplified form, the resolution improves as the maximum optical path difference increases:

$$\delta\tilde{\nu} \sim \frac{1}{x_{\max}}. \tag{13}$$

This explains why spectra recorded with larger mirror ranges should show sharper and better-separated water-vapour lines than spectra recorded with shorter mirror ranges. However, the practical line shape is also influenced by processing choices such as apodisation and zerofilling.

2.7 Zerofilling, Apodisation, and Phase Correction

Because the measured interferogram is finite, the calculated spectrum can show artificial oscillations around spectral lines. This is known as the leakage effect. It occurs because abruptly cutting off the interferogram corresponds mathematically to multiplying the ideal infinite interferogram by a rectangular window. To reduce this effect, the interferogram can be multiplied by an apodisation function before Fourier transformation.

Apodisation suppresses side lobes and ringing artifacts, but it also broadens the spectral lines. Therefore, it improves the visual smoothness of the spectrum at the cost of spectral resolution. Strong apodisation, such as a Blackman window, reduces leakage more strongly but gives broader peaks.

Weak or no apodisation preserves sharper features but leaves stronger oscillatory artifacts.

Zerofilling means extending the interferogram by adding zeros before the Fourier transform. This increases the number of calculated spectral points and makes the spectrum appear smoother. However, zerofilling does not add new physical information and therefore does not truly improve the resolution. It only improves the sampling density of the calculated spectrum.

If the measured interferogram is not perfectly symmetric around zero path difference, the Fourier transform gives a complex spectrum. A phase correction is then required to obtain the physically meaningful real-valued spectrum. In the practical evaluation, the amplitude or phase-corrected spectrum is used for the spectral analysis.

2.8 Molecular Model and Infrared Activity of HCl

The rotation-vibration spectrum of HCl is described using the diatomic molecule model. The molecule is treated as two point masses, connected by a bond. These two masses can rotate around their centre of mass and vibrate against each other. The centre-of-mass condition is

$$m_1 r_1 = m_2 r_2, \tag{14}$$

where m_1 and m_2 are the atomic masses and r_1, r_2 are their distances from the centre of mass. The reduced mass is defined as

$$\mu = \frac{m_1 m_2}{m_1 + m_2}. \tag{15}$$

The moment of inertia of the molecule is then

$$I = \mu r_0^2, \tag{16}$$

where r_0 is the bond length. The HCl molecule is infrared active because its vibration changes the electric dipole moment. Therefore, infrared radiation can excite vibrational and rotational-vibrational transitions, producing absorption lines in the measured spectrum.

2.9 Rotational Motion of a Diatomic Molecule

For a rigid diatomic molecule, the allowed rotational energy levels are quantized. The rotational energy is

$$E_J = hcBJ(J+1), \qquad J = 0, 1, 2, \ldots \tag{17}$$

where J is the rotational quantum number and B is the rotational constant. The rotational constant is related to the moment of inertia by

$$B = \frac{h}{8\pi^2 cI}. \tag{18}$$

Equivalently, using $I = \mu r_0^2$,

$$B = \frac{h}{8\pi^2 c\mu r_0^2}. \tag{19}$$

The rotational term value is

$$F(J) = \frac{E_J}{hc} = BJ(J+1). \tag{20}$$

For pure rotational transitions, the selection rule is

$$\Delta J = \pm 1. \tag{21}$$

In the present experiment, however, pure rotational transitions are not measured directly. Instead, rotational structure is observed together with the vibrational transition in the mid-infrared HCl spectrum.

2.10 Vibrational Motion and Force Constant

The vibration of a diatomic molecule can first be approximated as a harmonic oscillator. In this model, the two atoms oscillate around their equilibrium separation, and the restoring force is proportional to the displacement. The classical vibration wavenumber is

$$\tilde{\nu}_s = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}, \tag{22}$$

where k is the molecular force constant. Rearranging gives

$$k = \mu\left(2\pi c\tilde{\nu}_s\right)^2. \tag{23}$$

In the quantum-mechanical harmonic oscillator, the vibrational energy levels are

$$E_n = hc\tilde{\nu}_s\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots \tag{24}$$

where n is the vibrational quantum number. The allowed vibrational transitions satisfy

$$\Delta n = \pm 1. \tag{25}$$

For an ideal harmonic oscillator, all adjacent vibrational levels are equally spaced, so only one fundamental vibrational transition would be expected. In the measured HCl spectrum, this vibrational transition is accompanied by rotational transitions, producing the observed rotation-vibration band.

2.11 Rotation-Vibration Spectrum of HCl

When molecular rotation and vibration are considered together, the energy term of the rotating harmonic oscillator is

$$T(n, J) = BJ(J+1) + \tilde{\nu}_s\left(n + \frac{1}{2}\right). \tag{26}$$

For infrared rotation-vibration absorption, the relevant selection rules are

$$\Delta n = +1, \qquad \Delta J = \pm 1. \tag{27}$$

The branch with $\Delta J = +1$ is called the R branch. Its line positions in the rigid-rotator approximation are

$$\tilde{\nu}_R(J) = \tilde{\nu}_s + 2B(J+1), \qquad J = 0, 1, 2, \ldots \tag{28}$$

The branch with $\Delta J = -1$ is called the P branch. Its line positions are

$$\tilde{\nu}_P(J) = \tilde{\nu}_s - 2BJ, \qquad J = 1, 2, 3, \ldots \tag{29}$$

There is no central Q-branch line in this case because $\Delta J = 0$ is forbidden. Therefore, no absorption line appears exactly at the pure vibrational transition wavenumber $\tilde{\nu}_s$. Instead, $\tilde{\nu}_s$ can be obtained from neighbouring P- and R-branch lines:

$$\tilde{\nu}_s = \frac{\tilde{\nu}_R(0) + \tilde{\nu}_P(1)}{2}. \tag{30}$$

The rotational constant can be obtained from

$$B = \frac{\tilde{\nu}_R(0) - \tilde{\nu}_P(1)}{4}. \tag{31}$$

Once B is known, the moment of inertia and bond length can be determined from

$$I = \frac{h}{8\pi^2 c B}, \tag{32}$$

and

$$r_0 = \sqrt{\frac{I}{\mu}}. \tag{33}$$

These relations are the basis for extracting molecular parameters from the measured HCl absorption lines.

2.12 Non-Rigid Rotor, Isotope Splitting, and Intensity Distribution

The rigid-rotator approximation is not fully exact because a real rotating molecule is not perfectly rigid. At higher rotational quantum numbers, centrifugal effects slightly increase the bond length, increasing the moment of inertia and reducing the effective rotational spacing. This effect is described using the centrifugal distortion constant D:

$$F(J) = BJ(J+1) - DJ^2(J+1)^2. \tag{34}$$

If the rotational constant also depends on the vibrational state, the rotational term becomes

$$F_n(J) = B_n J(J+1) - D_n J^2(J+1)^2. \tag{35}$$

For practical evaluation, the measured line positions can be described using the running index i, where negative values correspond to the P branch and positive values to the R branch. In the approximation used for the experiment,

$$\tilde{\nu}(i) = \tilde{\nu}_s + (B_1 + B_0)i + (B_1 - B_0)i^2 - 2Di^3. \tag{36}$$

This relation allows the vibration-dependent rotational constants B_0 and B_1 and the centrifugal distortion constant D to be determined from the measured absorption-line positions.

The HCl spectrum also shows isotope splitting because chlorine occurs mainly as ^{35}Cl and ^{37}Cl. The two isotopologues, H^{35}Cl and H^{37}Cl, have different reduced masses. Since both the vibrational wavenumber and the rotational constant depend on the reduced mass, their absorption lines appear at slightly different positions. The isotope lines should therefore be assigned and evaluated separately.

The intensity distribution of the rotation-vibration lines is mainly determined by the thermal population of the lower rotational states. At room temperature, most molecules are in the vibrational ground state, but many rotational states are thermally populated. The population of a rotational state is proportional to the degeneracy factor $(2J+1)$ and the Boltzmann factor,

$$N(J) \propto (2J+1) \exp\left[-\frac{hcBJ(J+1)}{k_B T}\right]. \tag{37}$$

Using this population factor, the absorption coefficient can be written in the form

$$\alpha(J) = C\tilde{\nu}(J)(2J+1) \exp\left[-\frac{hcBJ(J+1)}{k_B T}\right], \tag{38}$$

where C is a proportionality constant. In Task 1.6, this expression is used to model the intensities of the absorption minima in the P branch, with the temperature T treated as a free fit parameter.

2.13 Interference Method for the Cuvette Gap Width

An empty cuvette can produce interference fringes because infrared light is reflected multiple times between the inner surfaces of the cuvette windows. The reflected beams remain coherent and can interfere constructively or destructively. In the transmission spectrum, this appears as a periodic modulation of the intensity.

For normal incidence and an air gap with refractive index $n_0 \approx 1$, the optical path difference between neighbouring reflected beams is

$$\Delta = 2n_0 d, \tag{39}$$

where d is the gap width. Transmission maxima occur when the optical path difference is an integer multiple of the wavelength:

$$2n_0 d = N\lambda, \qquad N = 0, 1, 2, \ldots \tag{40}$$

Using $\tilde{\nu} = 1/\lambda$, this can be written as

$$N = 2n_0 d\tilde{\nu}. \tag{41}$$

For two extrema with orders N_1 and N_2 at wavenumbers $\tilde{\nu}_1$ and $\tilde{\nu}_2$, subtraction gives

$$N_1 - N_2 = 2n_0 d\,(\tilde{\nu}_1 - \tilde{\nu}_2)\,. \tag{42}$$

Since the gap is filled with air, $n_0 \approx 1$, and the gap width is

$$d = \frac{N_1 - N_2}{2\,(\tilde{\nu}_1 - \tilde{\nu}_2)}. \tag{43}$$

In the evaluation, the order difference ΔN is plotted against the wavenumber difference $\Delta\tilde{\nu}$. The slope of the linear regression is

$$m = 2d, \tag{44}$$

so that the cuvette gap width is obtained from

$$d = \frac{m}{2}. \tag{45}$$

This method is used in Task 1.4 to determine the cuvette gap width and its uncertainty from the measured interference pattern.

3. Experimental Setup and Methods

3.1 FTIR Spectrometer

All measurements were carried out using the PerkinElmer Spectrum 100 Fourier-transform infrared spectrometer. The instrument operates in the mid-infrared spectral range and is based on a modified Michelson interferometer. Instead of measuring the spectrum directly by scanning individual wavelengths, the spectrometer records an interferogram as a function of optical path difference. The corresponding spectrum is then obtained by Fourier transformation.

The infrared radiation is generated by a glowbar source, consisting of a doped silicon carbide element heated electrically. This source emits a broad thermal spectrum with high intensity in the mid-infrared range. Inside the interferometer, the radiation is split into two beams, which travel different optical paths and are recombined before passing through the sample area and reaching the detector. The optical path difference is controlled by the interferometer mechanism, while a HeNe laser is used internally as a precise reference for the mirror/path-position calibration [1].

The spectra were recorded in transmission geometry. For each sample or optical component, a reference spectrum was recorded either without the sample in the beam path or using the corresponding background measurement. The measured sample spectrum was then compared with the reference spectrum in order to obtain the transmission spectrum. During the experiment, the sample compartment was kept closed except when changing samples, as recommended in the lab manual [1].

3.2 Samples and Optical Components

Several samples and optical components were used for the different parts of the experiment. Water vapour in the laboratory air and a polystyrene foil were used for checking the wavenumber calibration of the spectrometer. The water-vapour spectrum was also used for the interferogram measurements, since the known water-vapour absorption bands allow the Fourier-transformed spectra to be calibrated.

Glass and NaCl blocking filters were used to investigate the residual detected signal in spectral regions where the transmission should be strongly suppressed. An empty NaCl cuvette was used for the interference measurement, where multiple reflections between the cuvette windows generate interference fringes. Finally, a sealed HCl gas cuvette was used for the rotation-vibration spectrum. The HCl cuvette was not opened during the experiment. The NaCl components were handled carefully because hygroscopic materials can be damaged by moisture, and the polystyrene foil was handled carefully to avoid mechanical stress, in accordance with the precautions stated in the manual [1].

Table 1: Overview of samples and optical components used in the experiment.

Component	Purpose	Related task
Water vapour	Wavenumber calibration and interferogram analysis	1.1, 1.2
Polystyrene foil	Wavenumber calibration	1.1
Glass filter	Blocking-range measurement	1.3
NaCl filter	Blocking-range measurement	1.3
Empty NaCl cuvette	Interference fringes and gap-width determination	1.4
HCl gas cuvette	Rotation-vibration spectrum and intensity analysis	1.5, 1.6

3.3 General Measurement Workflow

For each measurement, an overview spectrum was first recorded in order to identify the relevant spectral region and to choose suitable measurement parameters. After this, the wavenumber range, spectral resolution, and number of scans were selected according to the requirements of the individual task. The main spectral range used for the calibration and filter measurements was from 200/cm to 4000/cm. Multiple scans were averaged in order to improve the signal-to-noise ratio.

For the wavenumber calibration, spectra of water vapour and polystyrene were recorded at several nominal resolutions. The positions of characteristic absorption bands were determined and compared with the reference values listed in the lab manual. The deviation of each band was calculated as the difference between the measured and reference wavenumber.

For the interferogram analysis, water-vapour interferograms were recorded using several different mirror-position ranges. The measured interferograms were Fourier transformed to obtain spectra. Since the Fourier-transformed

frequency axis was initially not directly calibrated in wavenumber units, known water-vapour absorption bands were used to convert the FFT frequency axis into a physical wavenumber axis. Different zerofilling factors and apodisation functions were then applied in order to compare their influence on the spectral appearance and resolution.

For the blocking-filter measurements, a reference spectrum without the filter was recorded first. Afterwards, the glass and NaCl filters were inserted separately into the beam path and measured under the same general conditions. The transmittance was calculated by dividing the filter spectrum by the reference spectrum. The blocking regions were then identified from the spectral intervals in which the transmission was strongly suppressed.

For the cuvette-gap measurement, the transmission spectrum of the empty NaCl cuvette was recorded. The interference fringes in the spectrum were analysed by selecting a sequence of clearly visible transmission minima. These minima were assigned relative fringe numbers and plotted against their corresponding wavenumber positions. A linear regression was then used to determine the gap width from the slope.

For the HCl analysis, the provided HCl spectrum and the corresponding background spectrum were used. After background correction, the absorption minima in the P and R branches were assigned to the two isotopologues $H^{35}Cl$ and $H^{37}Cl$. The assigned line positions formed the basis for determining the molecular constants and isotope splitting. The P-branch region was then used for the intensity analysis, where the absorption minima were background-corrected and fitted using the theoretical absorption-coefficient model.

3.4 Data Processing Workflow

The recorded and provided data were processed using external data-analysis tools after export from the spectrometer software. For the calibration spectra, absorption minima were identified and their positions were compared with the tabulated reference bands for water vapour and polystyrene. The wavenumber deviation was calculated using

$$\Delta\tilde{\nu} = \tilde{\nu}_{\mathrm{meas}} - \tilde{\nu}_{\mathrm{ref}}. \tag{46}$$

For the interferogram measurements, the raw interferograms were Fourier transformed. When required, the interferograms were first modified by zerofilling or by applying an apodisation window. The positive-frequency part of the resulting amplitude spectrum was used for further analysis. The FFT frequency axis was calibrated by assigning measured water-vapour features to their known reference wavenumbers.

For the filter measurements, the transmittance was calculated from the ratio of the filter spectrum to the reference spectrum,

$$T(\tilde{\nu}) = \frac{I_{\mathrm{filter}}(\tilde{\nu})}{I_{\mathrm{ref}}(\tilde{\nu})}. \tag{47}$$

The mean residual signal in the blocking range was then determined from the selected spectral interval. No interpretation of the residual signal was made at this stage; the values were evaluated later in the results and discussion section.

For the cuvette analysis, selected interference minima were assigned relative fringe orders. The relation between relative fringe number and wavenumber was fitted with a straight line. Since the slope is proportional to twice the cuvette gap width, the gap width was obtained from the fitted slope.

For the HCl spectrum, the background-corrected transmission spectrum was used to determine the absorption-line positions. The P- and R-branch lines were assigned separately for $H^{35}Cl$ and $H^{37}Cl$. From these line positions, the pure vibrational transition wavenumber, rotational constants, force constant, moment of inertia, bond length, centrifugal distortion, and isotope splitting were calculated using the equations introduced in the theoretical background. For the isotope-abundance and intensity analysis, a smooth baseline was fitted through the P-branch region after masking the absorption minima. The spectrum was divided by this baseline, and the corrected absorption depths were used for the Lambert-Beer and temperature-fit evaluations.

4. Results and Discussion

4.1 Wavenumber Scale Calibration Using Water Vapour and Polystyrene

4.1.1 Aim of the Measurement

The aim of this measurement was to verify the calibration of the wavenumber scale of the PerkinElmer Spectrum 100 FTIR spectrometer. This was done by recording spectra of water vapour and a polystyrene film and comparing the experimentally observed absorption-band positions with the known calibration band positions. The deviations between the measured and reference wavenumbers were then determined and plotted in order to check whether the spectrometer showed a systematic shift or whether the wavenumber scale was sufficiently well calibrated.

Although the task only requires the calibration check itself, the spectra were recorded at several spectral resolutions in order to observe how the resolution influences the visibility and accuracy of the absorption bands.

4.1.2 Measurement Procedure

For the calibration measurement, transmission spectra were recorded with the Spectrum 100 FTIR spectrometer over the wavenumber range from 200 cm^{-1} to 4000 cm^{-1}. The spectra were measured in energy units, EGY. For each measurement, 8 scans were averaged in order to improve the signal-to-noise ratio.

First, the water-vapour spectrum was measured with the sample compartment empty, so that the absorption bands originated from the water vapour present in the laboratory air. During the measurement, the relative humidity was approximately 28% and the room temperature was 13 °C. The measurement was repeated for the spectral resolutions 2 cm^{-1}, 4 cm^{-1}, 8 cm^{-1}, and 32 cm^{-1}.

The same measurement settings were then used for the polystyrene calibration sample. The polystyrene film was placed carefully in the sample holder and transmission spectra were recorded over the same wavenumber range, with the same number of scans and the same set of spectral resolutions. The positions of the characteristic polystyrene absorption bands were extracted from the measured spectra and compared with the corresponding reference values.

For both water vapour and polystyrene, the deviation of each measured peak position from its reference value was calculated as the difference between the experimentally determined wavenumber and the reference wavenumber. In symbolic form this is given by

$$\Delta\tilde{\nu} = \tilde{\nu}_{\mathrm{meas}} - \tilde{\nu}_{\mathrm{ref}}, \tag{48}$$

where $\tilde{\nu}_{\mathrm{meas}}$ is the experimentally determined peak position and $\tilde{\nu}_{\mathrm{ref}}$ is the corresponding reference wavenumber. These deviations were plotted as a function of the reference wavenumber. The resulting plots were used to evaluate whether the wavenumber scale of the spectrometer was correctly calibrated or whether a systematic offset was present.

4.1.3 Results

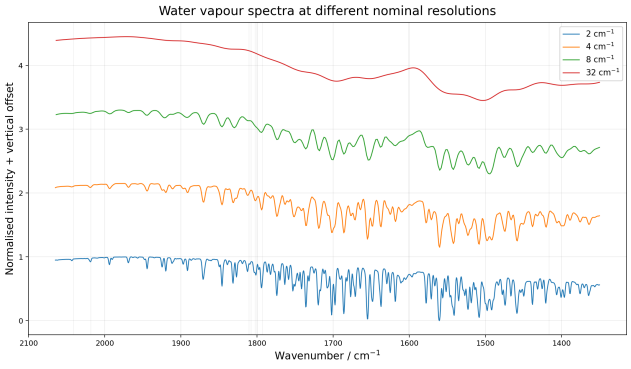


Figure 1: Water vapour spectra

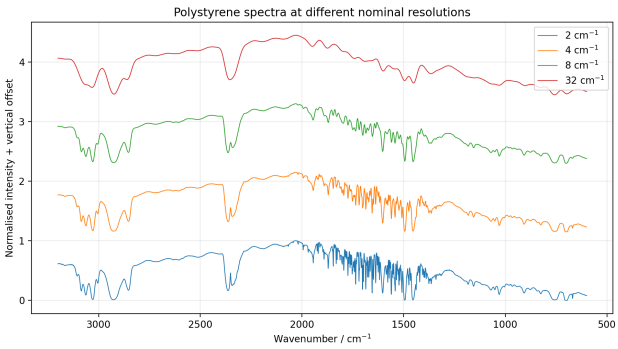


Figure 2: Polystyrene spectra

Table 2: Calibration-band comparison for the 2 1/cm resolution spectra.

Band source	Reference $\tilde{\nu}_{\mathrm{ref}}$ / 1/cm	Measured $\tilde{\nu}_{\mathrm{meas}}$ / 1/cm	Deviation / 1/cm
Water vapour	1416.08	1420.62	+4.54
Water vapour	1601.20	1604.77	+3.57
Water vapour	1792.65	1794.15	+1.50
Water vapour	1799.61	1795.00	-4.61
Water vapour	1802.47	1801.57	-0.90
Water vapour	1807.70	1803.40	-4.30
Water vapour	1810.62	1812.53	+1.91
Water vapour	2018.32	2018.96	+0.64
Water vapour	2041.30	2043.36	+2.06
Polystyrene	699.45	699.36	-0.09
Polystyrene	756.58	757.85	+1.27
Polystyrene	842.00	841.38	-0.62
Polystyrene	906.80	907.47	+0.67
Polystyrene	965.70	965.01	-0.69
Polystyrene	1028.30	1029.49	+1.19
Polystyrene	1069.10	1070.52	+1.42
Polystyrene	1154.60	1155.74	+1.14
Polystyrene	1368.50	1364.31	-4.19
Polystyrene	1449.70	1453.91	+4.21
Polystyrene	1542.20	1541.11	-1.09
Polystyrene	1583.10	1578.50	-4.60
Polystyrene	1601.40	1603.07	+1.67
Polystyrene	1803.80	1801.61	-2.19
Polystyrene	1945.20	1944.46	-0.74
Polystyrene	2850.00	2853.07	+3.07
Polystyrene	2920.90	2925.50	+4.60
Polystyrene	3001.40	3004.23	+2.83
Polystyrene	3026.40	3028.84	+2.44
Polystyrene	3060.00	3063.04	+3.04
Polystyrene	3082.20	3085.28	+3.08

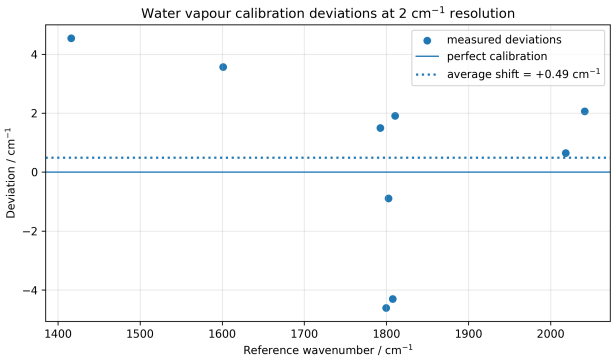


Figure 3: Deviation of measured calibration-band position for water vapour.

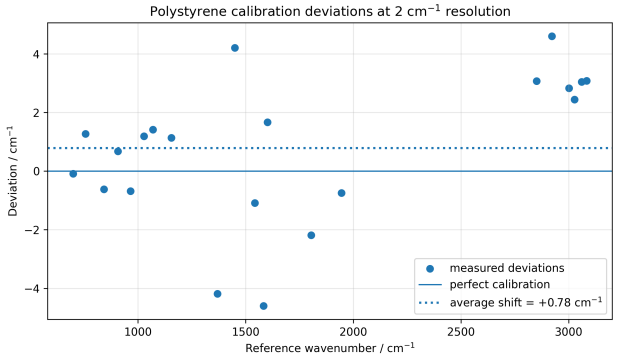


Figure 4: Deviation of measured calibration-band position for Polystyrene.

4.1.4 Discussion

The calibration of the wavenumber scale was checked by comparing the measured positions of the water-vapour and polystyrene absorption bands with their reference values. The deviations were calculated as

$$\Delta\tilde{\nu} = \tilde{\nu}_{\text{meas}} - \tilde{\nu}_{\text{ref}}, \tag{49}$$

where a positive value indicates a measured peak shifted to a higher wavenumber than the reference, and a negative value indicates a shift to a lower wavenumber.

For the water-vapour spectrum at 2 cm^{-1} resolution, the mean deviation was

$$\overline{\Delta\tilde{\nu}}_{\text{water}} = +0.49\ \text{cm}^{-1}, \tag{50}$$

and for the polystyrene spectrum it was

$$\overline{\Delta\tilde{\nu}}_{\text{polystyrene}} = +0.78\ \text{cm}^{-1}. \tag{51}$$

Both values are small compared with the full spectral range of 200 to 4000 cm^{-1}. The deviations fluctuate around zero with no strong systematic trend, indicating that the Spectrum 100 wavenumber scale was well calibrated. The small positive average shifts are not significant enough to indicate a serious calibration error. The remaining scatter can be attributed to finite spectral resolution, discrete data-point spacing, peak-fitting uncertainty, noise, and overlap between neighbouring absorption bands.

4.2 Water Vapour Interferograms, Fourier Transformation, and Spectral Resolution

4.2.1 Aim of the Measurement

The aim of this measurement was to study how the interferogram length influences the spectral resolution of a Fourier-transform infrared spectrum. For this purpose, water-vapour interferograms were recorded with different mirror ranges of the Spectrum 100 FTIR spectrometer. The spectra were obtained by Fourier transformation, calibrated using known water-vapour absorption bands, and then used to investigate the effects of zerofilling and apodisation.

4.2.2 Measurement Procedure

The water-vapour interferograms were recorded with the sample compartment empty, so that the observed absorption bands originated from the water vapour in the laboratory air. Measurements were taken for different mirror-position ranges: −500 to 500, −1000 to 1000, −4000 to 4000, −8000 to 8000, and −8000 to 32000 data-point units.

Each interferogram was Fourier transformed, and only the positive-frequency part of the resulting amplitude spectrum was used for the analysis. The initially uncalibrated FFT frequency axis was converted into a wavenumber axis by assigning measured water-vapour peak positions to their known reference values and applying a linear calibration. In addition, different zerofilling factors and apodisation functions were applied before the Fourier transformation in order to compare their influence on the calculated spectra.

4.2.3 Interferogram Data Calibration

The recorded interferograms are shown in Fig. 5. Increasing the mirror range increases the maximum optical path difference. Since closely spaced spectral lines produce slowly varying beat patterns in the interferogram, a longer interferogram contains more information about these components and therefore leads to a higher spectral resolution.

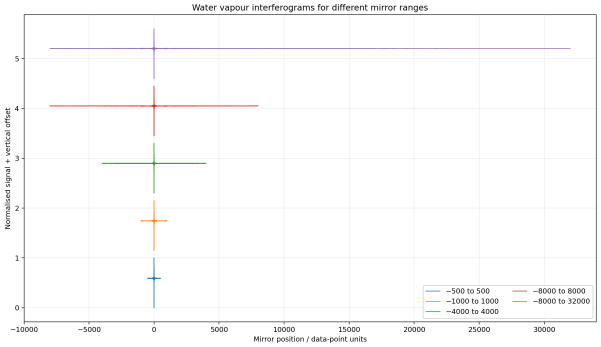


Figure 5: Water-vapour interferograms recorded for different mirror-position ranges. The curves are normalised and shifted vertically for clarity.

For the calibration, the largest-range interferogram was Fourier transformed and the measured water-vapour peak positions were assigned to the corresponding reference bands. The FFT frequency f was converted to the wavenumber scale using the linear relation

$$\tilde{\nu} = af + b.$$

The fit yielded

$$a = (15831.5 \pm 30.7)\ \text{cm}^{-1}, \qquad b = (-2.77 \pm 3.48)\ \text{cm}^{-1}.$$

Since the slope is related to the optical path step size by $a = 1/\Delta x$, the step size is

$$\Delta x = \frac{1}{a} = (6.3165{\times}10^{-5} {\pm} 1.2{\times}10^{-7})\ \text{cm} = (0.632 {\pm} 0.001)\ \mu\text{m}.$$

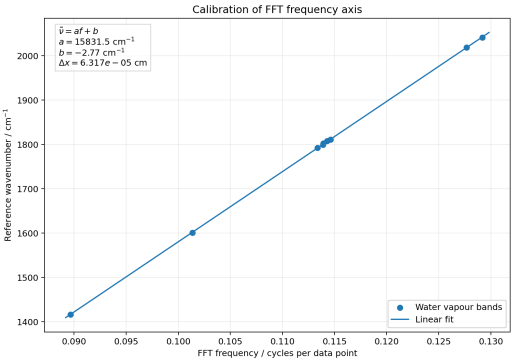


Figure 6: Linear calibration of the FFT frequency axis using water-vapour reference bands.

The root-mean-square residual of the calibration fit was 0.92 cm^{-1}, which is reasonable considering the partial overlap of several water-vapour bands.

After calibration, the Fourier-transformed spectra can be compared directly on the wavenumber axis, as shown in Fig. 7. The shorter mirror ranges show broad, unresolved absorption features, while the longer ranges resolve the individual water-vapour bands more clearly.

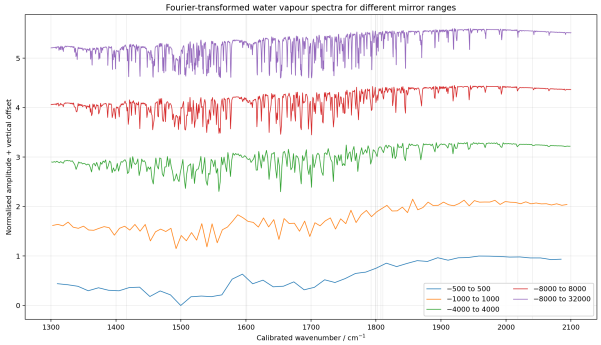


Figure 7: Fourier-transformed water-vapour spectra for different mirror ranges after calibration of the wavenumber axis. The grey vertical lines indicate reference water-vapour band positions.

4.2.4 Zerofilling

Zerofilling was performed by appending zeros to the measured interferogram before the Fourier transformation. This increases the number of calculated spectral points and therefore gives a smoother-looking spectrum. However, it does not add new physical information, so the true spectral resolution remains limited by the measured interferogram length.

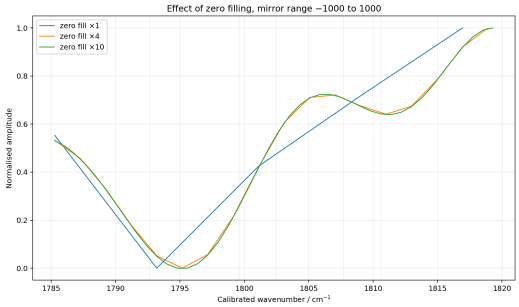


Figure 8: Effect of zerofilling for the −1000 to 1000 mirror range. The spectrum becomes smoother, but the close water-vapour bands remain unresolved.

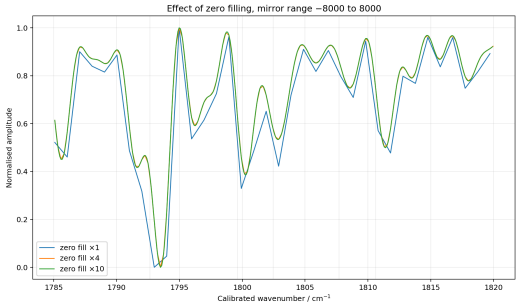


Figure 9: Effect of zerofilling for the −8000 to 8000 mirror range. The denser spectral sampling improves visual peak localisation, but the intrinsic linewidth is not reduced.

For the short mirror range, zerofilling mainly interpolates broad unresolved structures. For the longer mirror range, where the spectral features are already better resolved, zerofilling improves the visual smoothness and helps locate minima more precisely.

4.2.5 Apodisation

Apodisation was used to reduce leakage artifacts caused by the abrupt truncation of the finite interferogram. This was done by multiplying the interferogram with a window function before the Fourier transformation. While apodisation suppresses side lobes and oscillatory artifacts, it also broadens spectral lines and therefore reduces the effective resolution.

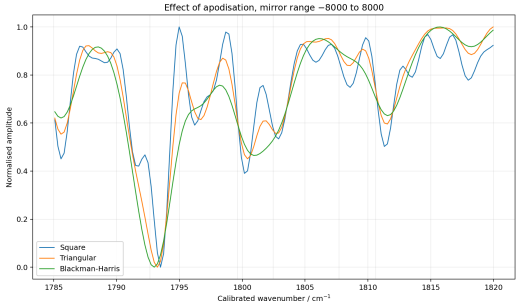


Figure 10: Effect of square, triangular, and Blackman-Harris apodisation for the −8000 to 8000 mirror range. The same zerofilling factor was used for all curves.



Figure 11: Effect of apodisation for the −1000 to 1000 mirror range. Strong apodisation broadens the already poorly resolved features.

For the longer mirror range, apodisation improves the baseline by suppressing spectral leakage, but the absorption minima become broader. For the short mirror range, the spectral features are already broad due to the limited interferogram length, so strong apodisation further worsens the separation of neighbouring bands.

Table 3: Detected water-vapour dip positions for mirror range -8000 to 8000.

Reference / 1/cm	Square / 1/cm	Triangular / 1/cm	Blackman-Harris / 1/cm
1416.080	1417.153	1419.255	1419.131
1601.200	1603.532	1603.656	1595.370
1792.650	1793.498	1793.251	1793.004
1799.610	1800.300	1800.424	1801.042
1802.470	1802.898	1800.424	1801.042
1807.700	1808.587	1808.463	1811.555
1810.620	1811.431	1811.431	1811.555
2018.320	2017.970	2018.094	2018.341
2041.300	2042.581	–	–

Table 4: Detected water-vapour dip positions for mirror range -1000 to 1000.

Reference / 1/cm	Square / 1/cm	Triangular / 1/cm	Blackman-Harris / 1/cm
1416.080	1420.364	1397.618	1397.618
1601.200	1617.170	1615.192	–
1792.650	1795.185	1793.207	–
1799.610	1795.185	1793.207	–
1802.470	1795.185	1793.207	1832.766
1807.700	1795.185	1793.207	1832.766
1810.620	1795.185	1793.207	1832.766
2018.320	2020.671	–	–
2041.300	–	–	–

Table 5: Summary of peak detection for different mirror ranges and apodisation functions.

Mirror range	Window	Detected peaks	Mean absolute deviation / 1/cm	Mean percentage error / %
-8000 to 8000	Square	9	0.97	0.056
-8000 to 8000	Triangular	8	1.36	0.084
-8000 to 8000	Blackman-Harris	8	2.11	0.128
-1000 to 1000	Square	8	8.10	0.469
-1000 to 1000	Triangular	7	11.51	0.692
-1000 to 1000	Blackman-Harris	4	23.99	1.399

The detected water-vapour dip positions show a clear dependence on both the mirror range and the apodisation function. For the longer mirror range (−8000 to 8000), all nine reference bands were detected with a mean absolute deviation of only 0.97 cm^{-1} using the square window, confirming that the longer interferogram provides sufficient resolution to locate the absorption bands accurately. Stronger apodisation increases the mean deviation to 1.36 cm^{-1} (triangular) and 2.11 cm^{-1} (Blackman–Harris), consistent with the expected line broadening.

For the shorter mirror range (−1000 to 1000), peak detection is significantly less reliable, with mean absolute deviations of 8.10 cm^{-1}, 11.51 cm^{-1}, and 23.99 cm^{-1} for the square, triangular, and Blackman–Harris windows, respectively. Several closely spaced bands around 1800 cm^{-1} are no longer resolved at this interferogram length. These results confirm that spectral resolution is primarily determined by the recorded interferogram length, while apodisation reduces artifacts at the cost of broader absorption lines.

4.2.6 Comparison between Zerofilling and Apodisation

Figure 12 compares a maximum-range spectrum with strong apodisation to a shorter-range spectrum with strong zerofilling. Zerofilling increases the sampling density of the calculated spectrum, but it cannot recover information that was not measured in the interferogram. In contrast, a longer interferogram improves the intrinsic resolving power because it records a larger optical path-difference interval. Apodisation modifies the line shape by reducing side lobes, but at the cost of broader spectral features.

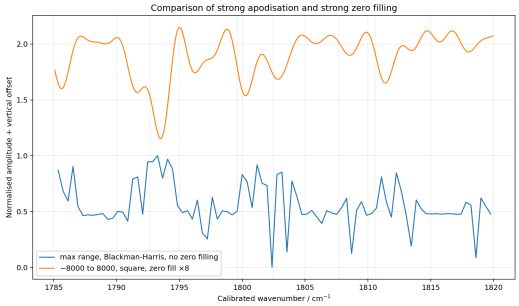


Figure 12: Comparison between strong apodisation of the maximum-range interferogram and strong zerofilling of the shorter −8000 to 8000 interferogram. The curves are normalised and shifted vertically for clarity.

Thus, zerofilling improves the visual representation of the spectrum, while apodisation reduces leakage artifacts. Neither method can replace the improvement in resolution obtained by recording a longer interferogram.

4.2.7 Spectral Resolution

The spectrometer covers the mid-infrared wavelength range from 2.5 μm to 50 μm. This corresponds to a maximum wavenumber of

$$\bar{\nu}_{\max} = 4000\,\mathrm{cm}^{-1}. \tag{52}$$

The interferometer step size is therefore given by

$$\Delta x = \frac{1}{2\bar{\nu}_{\max}} = \frac{1}{2 \cdot 4000} = 1.25 \times 10^{-4}\,\mathrm{cm}. \tag{53}$$

For a measurement with n sampling points, the theoretical spectral resolution becomes

$$\Delta\bar{\nu} = \frac{1}{n\,\Delta x} = \frac{8000}{n}\,\mathrm{cm}^{-1}. \tag{54}$$

4.3 Blocking Range of Glass and NaCl Filters

4.3.1 Aim of the Measurement

The aim of this measurement was to determine the residual detected signal strength in the blocking ranges of a glass filter and a NaCl blocking filter. Ideally, the transmittance in the blocking range should be close to zero. Therefore, any remaining signal in this region can be interpreted as residual transmission, detector background, noise, or artifacts from the measurement and data processing.

4.3.2 Measurement Procedure

First, a reference spectrum was recorded without any blocking filter in the beam path, followed by separate spectra of the glass and NaCl filters at spectral resolutions of 4 cm^{-1} and 8 cm^{-1}, each averaged over eight scans in the range 200 $\mathrm{cm}^{-1} \leq \bar{\nu} \leq 4000\ \mathrm{cm}^{-1}$. The transmittance was then calculated as

$$T(\bar{\nu}) = \frac{I_{\mathrm{filter}}(\bar{\nu})}{I_{\mathrm{ref}}(\bar{\nu})}, \tag{55}$$

where I_{filter} and I_{ref} are the intensities measured with and without the filter, respectively.

The blocking range was identified from the transmittance spectra as the region where T remains close to zero. This was taken as 250 $\mathrm{cm}^{-1} \leq \bar{\nu} \leq 2200\ \mathrm{cm}^{-1}$ for the glass filter and 250 $\mathrm{cm}^{-1} \leq \bar{\nu} \leq 500\ \mathrm{cm}^{-1}$ for the NaCl filter. The lower limit of 250 cm^{-1} was chosen to exclude the noisy, spike-dominated region at very low wavenumbers. After applying a median filter to suppress remaining spikes, the mean transmittance within the blocking range was calculated as

$$\overline{T}_{\mathrm{block}} = \frac{1}{N}\sum_{i=1}^{N} T_{\mathrm{filtered}}(\bar{\nu}_i), \tag{56}$$

where N is the number of data points in the selected range.

4.3.3 Results

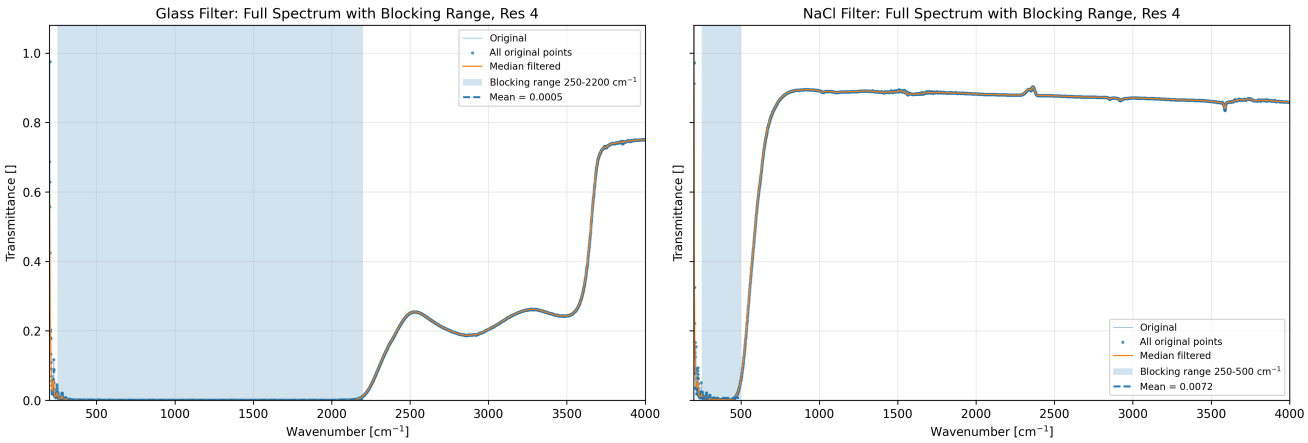


Figure 13: Transmission spectra used to identify the blocking ranges of the glass and NaCl filters

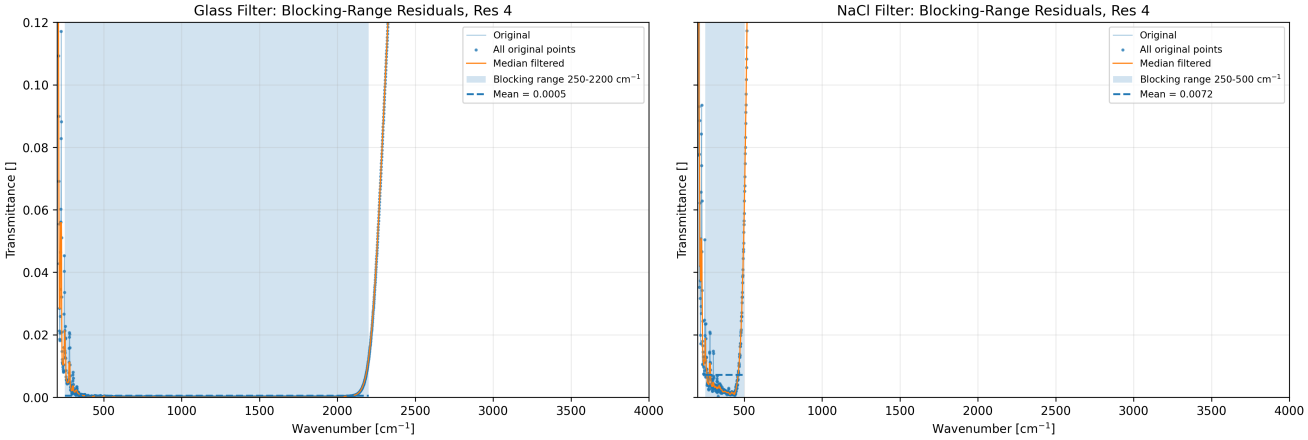


Figure 14: Blocking ranges of the glass and NaCl filters

Table 6: Comparison of the detected residual signal in the blocking ranges of the glass and NaCl filters at a resolution of 4 cm^{-1}.

Filter	Blocking range / 1/cm	Mean transmittance	Mean signal / %
Glass	250–2200	0.0005	0.05
NaCl	250–500	0.0072	0.72

4.3.4 Discussion

In the blocking ranges, the measured transmittance is expected to be close to zero, since the filters should suppress the incoming infrared radiation in these spectral regions. For the glass filter, the mean residual signal was found to be

$$\overline{T}_{\text{glass}} = 0.0005, \tag{57}$$

corresponding to approximately 0.05 %. This value is very small and indicates that the glass filter blocks the radiation very effectively in the selected range of 250–2200 cm^{-1}. The remaining signal is therefore most likely dominated by detector background, noise, and small artifacts from the Fourier transformation or background correction rather than by real transmission through the filter.

For the NaCl filter, the mean residual signal in the blocking range 250–500 cm^{-1} was

$$\overline{T}_{\text{NaCl}} = 0.0072, \tag{58}$$

corresponding to approximately 0.72 %. This is noticeably larger than the value obtained for the glass filter. The larger residual signal can be understood from the shape of the spectrum: the NaCl blocking region is much narrower and lies close to the steep transmission edge near 500 cm^{-1}, so the average is more strongly influenced by the rising transmittance near the cutoff. In addition, the low-wavenumber region contains stronger noise and spike-like artifacts, which can increase the apparent residual signal.

Both filters block effectively, with residual signals below 1 %, but the glass filter gives the lower value. Since the measured signals are close to the background level, the exact numerical values should not be interpreted as purely physical transmission, as they are affected by noise, background correction, and spike artifacts. The median filter applied to the data reduces the influence of these spikes and gives a more stable estimate of the residual signal.

4.4 Cuvette Gap Width from the Interference Method

4.4.1 Aim of the Measurement

The aim of this measurement was to determine the gap width of the NaCl cuvette using the interference method. Multiple reflections between the parallel cuvette windows produce periodic interference fringes whose spacing depends on the gap width.

4.4.2 Measurement Procedure

The transmission spectrum of the empty NaCl 1 cuvette was recorded with the Spectrum 100 FTIR spectrometer over the range

$$700\ \text{cm}^{-1} \text{ to } 4000\ \text{cm}^{-1},$$

using spectral resolutions of

$$1\ \text{cm}^{-1} \quad \text{and} \quad 4\ \text{cm}^{-1},$$

with 16 scans averaged for each spectrum.

Transmission minima in a region of clearly visible interference fringes were selected and assigned relative fringe numbers ΔN. The corresponding wavenumber positions $\tilde{\nu}_N$ were determined and plotted as ΔN versus $\tilde{\nu}_N$.

Using the interference condition, which predicts a linear relation with slope $2d$, the gap width d of the cuvette was determined from half of the fitted slope obtained by linear regression.

4.4.3 Linear Regression

$$\Delta N = 2d\,\Delta\tilde{\nu} \tag{59}$$

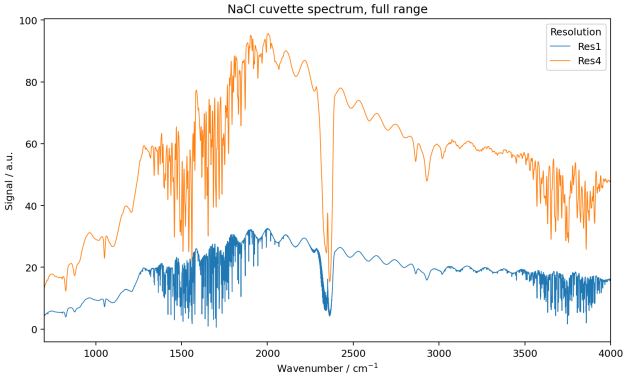


Figure 15: NaCl full range spectrum

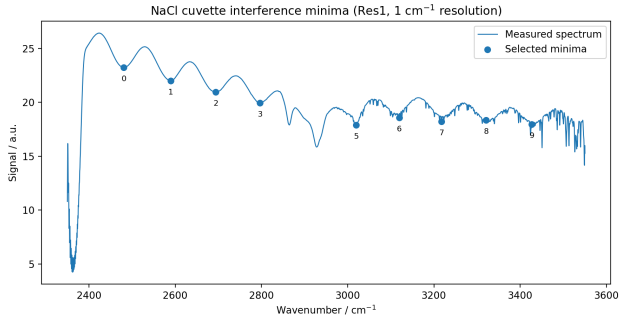


Figure 16: Interference minima

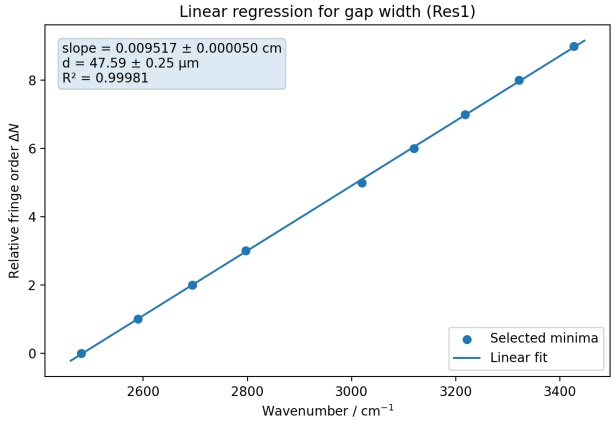


Figure 17: Linear Regression

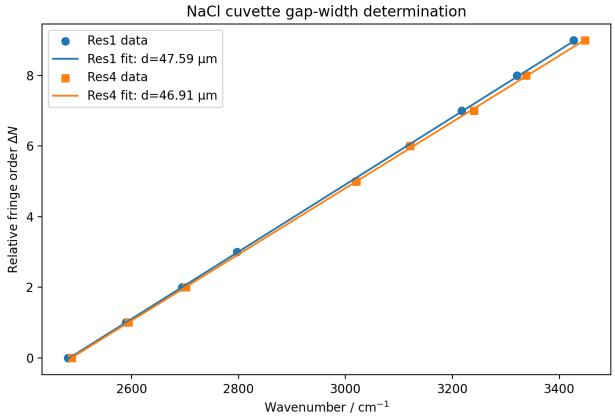


Figure 18: Linear regression combined

Table 7: Cuvette-gap regression results for the NaCl 1 cuvette.

Quantity	Resolution 1 cm^{-1}	Resolution 4 cm^{-1}
Slope $m = 2d$ / cm	9.517×10^{-3}	9.382×10^{-3}
Standard error of slope / cm	4.96×10^{-5}	3.73×10^{-5}
Gap width d / μm	47.59 ± 0.25	46.91 ± 0.19
Mean square error	1.74×10^{-3}	9.41×10^{-4}
R^2	0.99981	0.99991

4.4.4 Discussion

The linear fits show a very good agreement with the interference relation, as both resolutions give values of R^2 close to one. This confirms that the selected transmission minima follow an approximately linear dependence between the relative fringe order ΔN and the wavenumber $\tilde{\nu}_N$. The gap widths obtained from the two spectra are also in good agreement, with $d = (47.59 \pm 0.25)$ μm for the 1 cm^{-1} spectrum and $d = (46.91 \pm 0.19)$ μm for the 4 cm^{-1} spectrum.

The remaining deviations from the fit are mainly caused by uncertainty in selecting the exact positions of the interference minima. Additional uncertainty may come from noise, imperfect baseline correction, overlapping spectral structures, and slight non-parallelism of the cuvette windows. The minima were easier to identify than the maxima in the selected wavenumber region because they appeared sharper and more regularly spaced. Therefore, only transmission minima were used for the fringe numbering and linear regression.

4.5 Rotation-Vibration Spectrum of HCl

4.5.1 Aim of the Measurement

The aim of this task was to analyse the rotation-vibration spectrum of hydrogen chloride gas. The HCl spectrum and corresponding background spectrum were already provided as data files, so no additional measurement had to be performed. From the background-corrected HCl transmission spectrum, the spectral positions of the absorption lines in the P- and R-branches were determined for the two isotopologues $H^{35}Cl$ and $H^{37}Cl$.

These line positions were then used to calculate important molecular parameters, such as the pure vibrational transition wavenumber, rotational constant, force constant, moment of inertia, and bond length. In addition, the isotope splitting, vibration-dependent rotational constants, centrifugal distortion constant, and relative isotope abundance were determined from the spectrum.

4.5.2 Overview Spectrum and Branch Assignment

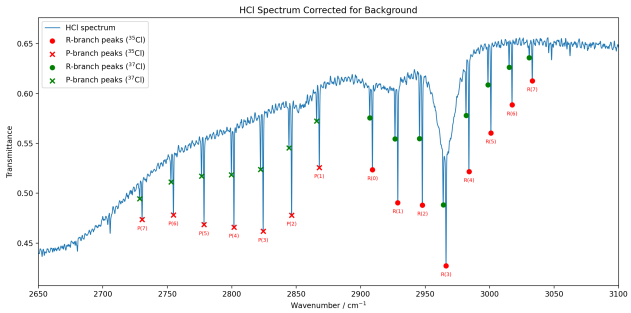


Figure 19: Full HCl rotation-vibration spectrum with labelled P and R branches

Table 8: Assigned HCl rotation-vibration absorption-band positions.

Transition	Index i	$\tilde{\nu}(^{35}Cl)$ / 1/cm	$\tilde{\nu}(^{37}Cl)$ / 1/cm	Splitting / 1/cm
P(7)	-7	2730.500	2728.625	1.875
P(6)	-6	2754.750	2752.875	1.875
P(5)	-5	2778.500	2776.500	2.000
P(4)	-4	2801.750	2799.750	2.000
P(3)	-3	2824.375	2822.375	2.000
P(2)	-2	2846.375	2844.375	2.000
P(1)	-1	2867.875	2865.875	2.000
R(0)	1	2909.125	2907.000	2.125
R(1)	2	2928.750	2926.625	2.125
R(2)	3	2947.875	2945.625	2.250
R(3)	4	2966.250	2964.000	2.250
R(4)	5	2984.000	2981.750	2.250
R(5)	6	3001.000	2998.750	2.250
R(6)	7	3017.375	3015.125	2.250
R(7)	8	3033.125	3030.750	2.375

Pure Vibrational Transition:

$$\tilde{\nu}_s = \frac{\tilde{\nu}_R(0) + \tilde{\nu}_P(1)}{2} \tag{60}$$

Rotational Constant:

$$B = \frac{\tilde{\nu}_R(0) - \tilde{\nu}_P(1)}{4} \tag{61}$$

Force Constant:

$$k = \mu\,(2\pi c\tilde{\nu}_s)^2 \tag{62}$$

Moment of Inertia and Bond Length:

$$I = \frac{h}{8\pi^2 cB} \tag{63}$$

$$r_0 = \sqrt{\frac{I}{\mu}} \tag{64}$$

Table 9: Spectroscopic constants and molecular parameters obtained from the HCl spectrum.

Quantity	$H^{35}Cl$	$H^{37}Cl$
$\tilde{\nu}_s$ / 1/cm	2888.50	2886.44
B / 1/cm	10.31	10.28
μ / kg	1.6267×10^{-27}	1.6291×10^{-27}
k / N/m	481.55	481.59
I / kg m^2	2.7144×10^{-47}	2.7227×10^{-47}
r_0 / m	1.2918×10^{-10}	1.2928×10^{-10}

Where:

$\tilde{\nu}_s$: Pure vibrational transition
B : Rotational constant
μ : Reduced mass
k : Force constant
I : Moment of inertia
r_0 : Bond length

4.5.3 *Quadratic curve fit to obtain D, B_0 and B_1*

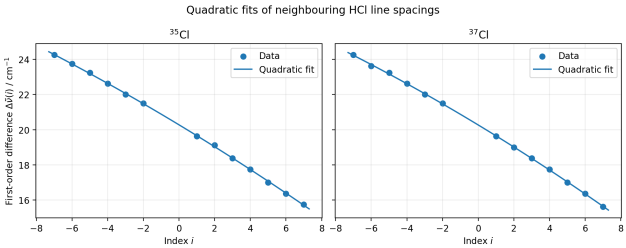


Figure 20: Quadratic fits of neighbouring HCl line spacing

Table 10: Quadratic-fit parameters obtained from the first-order differences of neighbouring HCl absorption lines.

Parameter	^{35}Cl	^{37}Cl
D / 1/cm	$(4.90 \pm 0.74) \times 10^{-4}$	$(5.67 \pm 0.68) \times 10^{-4}$
$B_1 - B_0$ / 1/cm	-0.3031 ± 0.0016	-0.3033 ± 0.0015
B_1 / 1/cm	10.1405 ± 0.0119	10.1362 ± 0.0109
B_0 / 1/cm	10.4436 ± 0.0120	10.4395 ± 0.0110
RMS residual / 1/cm	0.0450	0.0413

4.5.4 *Estimating Isotopic abundance using Lambert-Beer law*

To determine the isotopic abundance ratio, we first corrected the P-branch region of the HCl spectrum for a slowly varying background. The absorption minima belonging to the HCl isotope peaks were excluded from the baseline fit, so only the underlying continuum of the spectrum was used. Then a smooth polynomial baseline was fitted through these remaining points. The original spectrum was divided by this fitted baseline to obtain a normalized spectrum in which the depths of the absorption minima represent the relative absorption strengths of the two isotopologues. These relative peak depths were then compared to estimate the abundance ratio of ^{35}Cl and ^{37}Cl in the HCl gas sample.

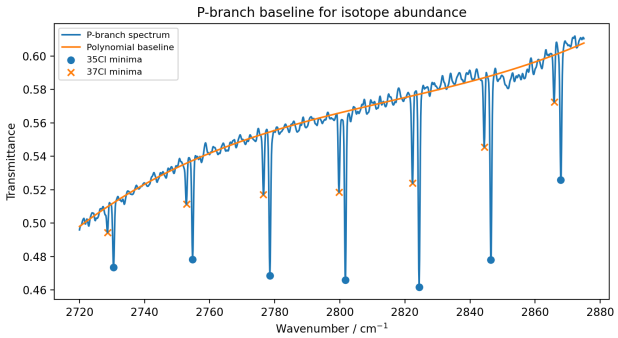


Figure 21: P-branch baseline for isotope abundance

Table 11: Isotopic abundance estimate from the P-branch using the Lambert–Beer law.

Isotope	Sum of absorbance depths	Abundance / %
^{35}Cl	1.1254	71.5
^{37}Cl	0.4490	28.5

4.5.5 *Discussion*

The background-corrected HCl spectrum shows the expected rotation-vibration structure, with a P-branch at lower wavenumbers and an R-branch at higher wavenumbers. The missing central line is expected because the transition with $\Delta J = 0$ is forbidden. Each absorption line appears as a doublet due to the two chlorine isotopes, Cl-35 and Cl-37.

The Cl-35 lines occur at slightly higher wavenumbers than the Cl-37 lines. This is physically reasonable because

Cl-37 gives a larger reduced mass, which lowers both the vibrational frequency and the rotational constant. The observed isotope splitting therefore has the expected direction.

The calculated molecular parameters are also reasonable. The pure vibrational transition is near 2888 cm^{-1}, the force constant is about 482 $N\,m^{-1}$, and the bond length is approximately 1.29×10^{-10} m, all of which are realistic for HCl. The moment of inertia is slightly larger for the Cl-37 isotopologue, as expected.

The quadratic fits show a slight decrease in line spacing with increasing rotational quantum number, which is consistent with centrifugal distortion. The obtained distortion constants are much smaller than the rotational constants, so the approximation $D \ll B$ is valid.

The isotope abundance analysis gave approximately 71.5% for Cl-35 and 28.5% for Cl-37. This agrees reasonably well with the natural chlorine abundances of about 75.8% for Cl-35 and 24.2% for Cl-37. The measured result therefore correctly shows that Cl-35 is the more abundant isotope. The deviation from the literature values is most likely caused by uncertainty in the baseline correction, overlap of neighbouring isotope peaks, finite spectral resolution, and the difficulty of determining the exact absorption depths.

Overall, the measured spectrum and calculated parameters are physically consistent with the expected rotation-vibration behaviour of HCl. The main uncertainty sources are peak assignment, finite resolution, overlapping isotope lines, noise, and baseline correction.

4.6 Intensity Modelling of the P-Branch Absorption Minima

4.6.1 Aim of the Model

The aim of this model was to describe the measured intensities of the HCl P-branch absorption minima using the theoretical absorption coefficient. The temperature was treated as a free fit parameter, so that the rotational population distribution could be compared with the experimental intensity distribution.

4.6.2 Background Correction

The HCl spectrum was corrected by estimating the slowly varying background in the P-branch region. The absorption minima were first masked, so that they did not influence the background fit. A smooth B-spline baseline was then fitted through the remaining points. Finally, the measured spectrum was divided by this baseline to obtain a normalized spectrum, in which the absorption minima could be compared more reliably.

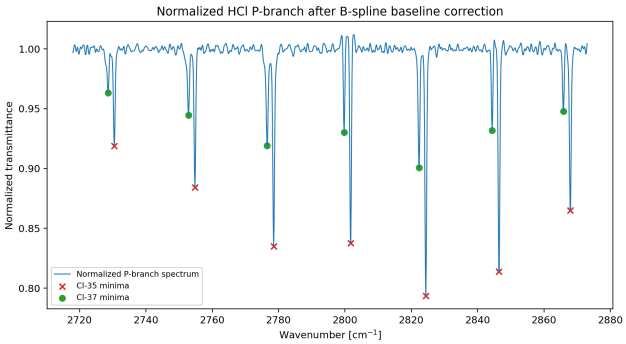


Figure 22: Normalized HCl P-Branch

4.6.3 Model Function

$$\alpha(J) = C\tilde{\nu}(J)(2J+1)\exp\left[-\frac{hcBJ(J+1)}{k_B T}\right] \tag{65}$$

$$I(J) = I_0 \exp(-\alpha d) = I_0 \exp\left(-c_4 \bar{\nu}(2J+1)\exp\left(-\frac{hcBJ(J+1)}{kT}\right)\right) \tag{66}$$

4.6.4 Fit Procedure

After background correction, the transmittance minima of the P-branch were extracted from the normalized HCl spectrum separately for Cl-35 and Cl-37. These minima were then fitted as a function of the rotational quantum number J using the Lambert-Beer law shown above.

In the fit, the theoretical absorption coefficient $\alpha(J)$ describes the dependence of the absorption strength on the rotational population of the molecule. The temperature T was treated as a free parameter, together with an overall scaling factor. The fitted value of T therefore gives the effective temperature obtained from the measured intensity distribution of the P-branch minima.

4.6.5 Fit Results

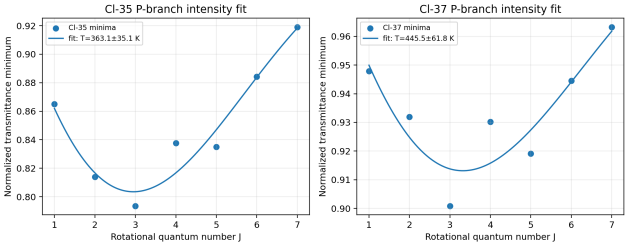


Figure 23: Fits for the normalized transmittance minima of the dips using the Absorption coefficient model

Table 12: P-branch normalized minima after masked B-spline baseline correction.

Isotope	J	Peak position / cm^{-1}	Normalized minimum
^{35}Cl	1	2867.875	0.8650
^{35}Cl	2	2846.375	0.8138
^{35}Cl	3	2824.375	0.7935
^{35}Cl	4	2801.750	0.8376
^{35}Cl	5	2778.500	0.8349
^{35}Cl	6	2754.750	0.8842
^{35}Cl	7	2730.500	0.9190
^{37}Cl	1	2865.875	0.9479
^{37}Cl	2	2844.375	0.9319
^{37}Cl	3	2822.375	0.9009
^{37}Cl	4	2799.750	0.9302
^{37}Cl	5	2776.500	0.9191
^{37}Cl	6	2752.875	0.9446
^{37}Cl	7	2728.625	0.9632

Table 13: Temperature fit results for the HCl P-branch using the theoretical absorption-coefficient model.

Isotope	Temperature / K	R^2	RMS residual
^{35}Cl	363.1 ± 35.1	0.937	0.0100
^{37}Cl	445.5 ± 61.8	0.800	0.0084

4.6.6 Discussion

The fitted temperatures were found to be $363.1 \pm 35.1\ K$ for ^{35}Cl and $445.5 \pm 61.8\ K$ for ^{37}Cl. Both values are higher than the expected room temperature of approximately $293\ K$. The ^{35}Cl result is still of the correct order of magnitude and relatively close to room temperature within a few standard deviations, while the ^{37}Cl result deviates more strongly and also has a larger uncertainty.

This difference is physically understandable because the temperature fit is very sensitive to the extracted absorption intensities. Small changes in the baseline correction can change the apparent depth of the absorption minima and therefore affect the fitted temperature. In addition, some absorption lines may be partially saturated, so their depths no longer scale linearly with the theoretical absorption coefficient. Overlap between neighbouring isotope lines can also distort the extracted peak depths, especially for the weaker ^{37}Cl lines. Finally, the finite spectral resolution broadens the absorption features and can make the true minima harder to determine accurately.

Therefore, the fitted temperatures should be interpreted as effective temperatures obtained from the intensity distribution rather than precise gas temperatures. The results are qualitatively reasonable because the fitted curves reproduce the general trend of the P-branch intensities, but the deviations from room temperature show that the intensity analysis is limited by baseline correction, peak overlap, saturation effects, and finite resolution.

5. Conclusion

In this experiment, Fourier-transform infrared spectroscopy was used to investigate instrumental calibration, interferogram processing, optical transmission effects, interference fringes, and the rotation-vibration spectrum of hydrogen chloride. The measurements and evaluations showed that the PerkinElmer Spectrum 100 FTIR spectrometer was suitable for resolving the relevant mid-infrared spectral features and for extracting physically meaningful molecular parameters from the measured spectra.

The wavenumber calibration was verified using water-vapour and polystyrene absorption bands. The mean deviations were small, with average shifts of $\Delta\tilde{\nu}_{\text{water}} = 0.49/\text{cm}$ for water vapour and $\Delta\tilde{\nu}_{\text{polystyrene}} = 0.78/\text{cm}$ for polystyrene. Since the deviations fluctuated around zero and showed no strong systematic trend, the wavenumber scale of the spectrometer can be considered sufficiently well calibrated for the later analysis of the HCl spectrum.

The analysis of water-vapour interferograms demonstrated the central role of interferogram length in FTIR spectroscopy. Longer mirror ranges produced better resolved spectra because they correspond to larger optical path differences and therefore higher spectral resolution. Zerofilling improved the visual smoothness and point density of the calculated spectra, but did not improve the true physical resolution. Apodisation reduced leakage artifacts and oscillatory side structures, but broadened the absorption lines and therefore reduced the effective resolution. The comparison confirmed that improved resolution can only be obtained by recording a sufficiently long interferogram, while zerofilling and apodisation mainly influence the numerical and visual appearance of the spectrum.

The blocking-filter measurements showed that both the glass and NaCl filters suppressed the transmitted infrared signal effectively in their blocking ranges. The residual signal was very small for the glass filter and somewhat larger for the NaCl filter, mainly because the selected NaCl blocking range lies close to a cutoff region and is more strongly affected by low-wavenumber noise. Overall, both filters behaved as expected, with the remaining detected signal being mainly attributable to background, noise, and processing artifacts rather than significant real transmission.

The interference measurement with the empty NaCl cuvette produced a clear periodic fringe pattern. By assigning relative fringe numbers to selected transmission minima and applying a linear regression, the gap width was determined as $d = (47.59 \pm 0.25)\ \mu\text{m}$ for the $1/\text{cm}$ spectrum and $d = (46.91 \pm 0.19)\ \mu\text{m}$ for the $4/\text{cm}$ spectrum. The close agreement between the two values and the high linearity of the regression show that the interference method gives a reliable estimate of the cuvette gap width.

The background-corrected HCl spectrum showed the expected rotation-vibration structure, with a P branch at lower

wavenumbers and an R branch at higher wavenumbers. The absence of a central Q branch is consistent with the selection rule $\Delta J = 0$ being forbidden. The absorption lines appeared as isotope doublets due to $H^{35}Cl$ and $H^{37}Cl$. The extracted pure vibrational transition wavenumbers were approximately $2888.50/\mathrm{cm}$ for $H^{35}Cl$ and $2886.44/\mathrm{cm}$ for $H^{37}Cl$, while the corresponding rotational constants were $B = 10.31/\mathrm{cm}$ and $B = 10.28/\mathrm{cm}$. From these values, realistic molecular parameters were obtained, including force constants of about $482\,\mathrm{N/m}$ and bond lengths of approximately $1.29 \times 10^{-10}\,\mathrm{m}$. The line-spacing analysis also showed a small centrifugal distortion effect, with distortion constants much smaller than the rotational constants, as expected.

The isotope abundance analysis gave approximately 71.5 % for ^{35}Cl and 28.5 % for ^{37}Cl. This is reasonably close to the natural chlorine isotope abundances and correctly identifies ^{35}Cl as the more abundant isotope. The remaining deviation is most likely caused by baseline uncertainty, overlap between neighbouring isotope lines, finite spectral resolution, and uncertainty in determining the exact absorption depths.

Finally, the P-branch intensity modelling reproduced the general shape of the measured absorption-minimum distribution. The fitted effective temperatures were higher than room temperature, especially for $H^{37}Cl$, so they should not be interpreted as precise gas temperatures. Instead, they show that the intensity model captures the qualitative rotational population trend, while the quantitative result is limited by baseline correction, partial saturation of absorption lines, overlap of isotope peaks, and finite spectral resolution.

Overall, the experiment successfully demonstrated how FTIR spectroscopy can be used not only to record infrared spectra, but also to extract detailed physical information from them. The calibration, interferogram-processing, filter, cuvette, and HCl analyses all gave physically consistent results. The main limitations were finite spectral resolution, peak-position uncertainty, noise, baseline correction, and overlap between neighbouring absorption lines. Despite these limitations, the measured spectra and derived parameters agree well with the expected behaviour of rotation-vibration spectra of a diatomic molecule.

References

[1] Volker Riede and Christian Kranert. *Rotation-Vibration Spectra of Molecules*. Experiment for the Advanced Physics Laboratory Course. Fakultät für Physik und Geowissenschaften, Universität Leipzig.

Appendix

A. Calibration Band Tables

Table 14: Calibration-band comparison for the water-vapour and polystyrene spectra measured at a nominal resolution of 2/cm. The deviation is defined as $\Delta\tilde{\nu} = \tilde{\nu}_{\mathrm{meas}} - \tilde{\nu}_{\mathrm{ref}}$.

Band source	$\tilde{\nu}_{\mathrm{ref}}$ / 1/cm	$\tilde{\nu}_{\mathrm{meas}}$ / 1/cm	Deviation / 1/cm
Water vapour	1416.08	1420.62	+4.54
Water vapour	1601.20	1604.77	+3.57
Water vapour	1792.65	1794.15	+1.50
Water vapour	1799.61	1795.00	-4.61
Water vapour	1802.47	1801.57	-0.90
Water vapour	1807.70	1803.40	-4.30
Water vapour	1810.62	1812.53	+1.91
Water vapour	2018.32	2018.96	+0.64
Water vapour	2041.30	2043.36	+2.06
Polystyrene	699.45	699.36	-0.09
Polystyrene	756.58	757.85	+1.27
Polystyrene	842.00	841.38	-0.62
Polystyrene	906.80	907.47	+0.67
Polystyrene	965.70	965.01	-0.69
Polystyrene	1028.30	1029.49	+1.19
Polystyrene	1069.10	1070.52	+1.42
Polystyrene	1154.60	1155.74	+1.14
Polystyrene	1368.50	1364.31	-4.19
Polystyrene	1449.70	1453.91	+4.21
Polystyrene	1542.20	1541.11	-1.09
Polystyrene	1583.10	1578.50	-4.60
Polystyrene	1601.40	1603.07	+1.67
Polystyrene	1803.80	1801.61	-2.19
Polystyrene	1945.20	1944.46	-0.74
Polystyrene	2850.00	2853.07	+3.07
Polystyrene	2920.90	2925.50	+4.60
Polystyrene	3001.40	3004.23	+2.83
Polystyrene	3026.40	3028.84	+2.44
Polystyrene	3060.00	3063.04	+3.04
Polystyrene	3082.20	3085.28	+3.08

B. Measurement Settings

Table 15: Summary of the main measurement settings used in the experiment.

Measurement	Settings	Comment
Wavenumber calibration	Range: 200/cm to 4000/cm; spectra measured in EGY mode; 8 scans averaged; nominal resolutions of 2/cm, 4/cm, 8/cm, and 32/cm.	Water vapour was measured with the sample compartment empty. Polystyrene foil was inserted into the sample holder.
Water-vapour interferograms	Mirror-position ranges: −500 to 500, −1000 to 1000, −4000 to 4000, −8000 to 8000, and −8000 to 32000 data-point units.	Used to investigate the influence of interferogram length, Fourier transformation, zerofilling, and apodisation.
Zerofilling comparison	Different zerofilling factors were applied before Fourier transformation.	Used to increase spectral point density without changing the true physical resolution.
Apodisation comparison	Square, triangular, and Blackman-Harris windows were compared.	Used to study the trade-off between leakage suppression and spectral broadening.

Measurement	Settings	Comment
Glass and NaCl blocking filters	Range: 200/cm to 4000/cm; resolutions of 4/cm and 8/cm; 8 scans averaged.	Used to determine residual detected signal in the blocking ranges.
NaCl cuvette interference	Range: 700/cm to 4000/cm; resolutions of 1/cm and 4/cm; 16 scans averaged.	Used to determine the cuvette gap width from interference fringes.
HCl rotation-vibration spectrum	Provided HCl spectrum and corresponding background spectrum; background correction applied before analysis.	Used for line assignment, molecular constants, isotope splitting, isotope abundance, and P-branch intensity modelling.

C. Full HCl Line Assignment

Table 16: Assigned HCl rotation-vibration absorption-band positions for $H^{35}Cl$ and $H^{37}Cl$.

Transition	Index i	$\tilde{\nu}(H^{35}Cl)$ / 1/cm	$\tilde{\nu}(H^{37}Cl)$ / 1/cm	Splitting / 1/cm
P(7)	-7	2730.500	2728.625	1.875
P(6)	-6	2754.750	2752.875	1.875
P(5)	-5	2778.500	2776.500	2.000
P(4)	-4	2801.750	2799.750	2.000
P(3)	-3	2824.375	2822.375	2.000
P(2)	-2	2846.375	2844.375	2.000
P(1)	-1	2867.875	2865.875	2.000
R(0)	1	2909.125	2907.000	2.125
R(1)	2	2928.750	2926.625	2.125
R(2)	3	2947.875	2945.625	2.250
R(3)	4	2966.250	2964.000	2.250
R(4)	5	2984.000	2981.750	2.250
R(5)	6	3001.000	2998.750	2.250
R(6)	7	3017.375	3015.125	2.250
R(7)	8	3033.125	3030.750	2.375

D. Summary of Evaluation Formulae

Table 17: Summary of the main formulae used for data evaluation and error propagation.

Quantity / Purpose	Formula	Used in
Calibration deviation	$\Delta\tilde{\nu} = \tilde{\nu}_{\mathrm{meas}} - \tilde{\nu}_{\mathrm{ref}}$	Wavenumber calibration
Mean calibration deviation	$\overline{\Delta\tilde{\nu}} = \frac{1}{N}\sum_{i=1}^{N}\Delta\tilde{\nu}_i$	Water vapour and polystyrene calibration
Mean absolute deviation	$\mathrm{MAD} = \frac{1}{N}\sum_{i=1}^{N}\lvert\Delta\tilde{\nu}_i\rvert$	Comparison of peak-position accuracy
FFT-axis calibration	$\tilde{\nu} = af + b$	Calibration of Fourier-transformed interferograms
Optical path step size	$\Delta x = \frac{1}{a}$	Interferogram data calibration

Quantity / Purpose	Formula	Used in
Uncertainty of optical path step size	$\sigma_{\Delta x} = \frac{\sigma_a}{a^2}$	Error propagation from FFT-axis fit
Theoretical spectral spacing	$\Delta\tilde{\nu} = \frac{1}{n\Delta x}$	Spectral-resolution estimate
Sampling step from maximum wavenumber	$\Delta x = \frac{1}{2\tilde{\nu}_{\max}}$	Resolution estimate for the MIR range
Filter transmittance	$T(\tilde{\nu}) = \frac{I_{\text{filter}}(\tilde{\nu})}{I_{\text{ref}}(\tilde{\nu})}$	Glass and NaCl blocking-filter measurements
Mean residual blocking signal	$T_{\text{block}} = \frac{1}{N}\sum_{i=1}^{N} T_{\text{filtered}}(\tilde{\nu}_i)$	Residual signal in blocking ranges
Cuvette interference relation	$\Delta N = 2d\,\Delta\tilde{\nu}$	Cuvette gap-width determination
Gap width from regression slope	$d = \frac{m}{2}$	Linear regression of interference minima
Uncertainty of gap width	$\sigma_d = \frac{\sigma_m}{2}$	Error propagation of cuvette gap width
Pure vibrational transition	$\tilde{\nu}_s = \frac{\tilde{\nu}_R(0) + \tilde{\nu}_P(1)}{2}$	HCl rotation-vibration spectrum
Rotational constant	$B = \frac{\tilde{\nu}_R(0) - \tilde{\nu}_P(1)}{4}$	HCl molecular constants
Force constant	$k = \mu\,(2\pi c\tilde{\nu}_s)^2$	HCl molecular bond strength
Moment of inertia	$I = \frac{h}{8\pi^2 cB}$	HCl molecular geometry
Bond length	$r_0 = \sqrt{\frac{I}{\mu}}$	HCl molecular geometry
Absorbance from normalized transmittance	$A_i = -\ln(T_i)$	Isotope-abundance estimate
Relative isotope abundance	$p_j = \frac{\sum_i A_{j,i}}{\sum_i A_{^{35}\text{Cl},i} + \sum_i A_{^{37}\text{Cl},i}} \times 100\%$	Lambert-Beer isotope analysis

Quantity / Purpose	Formula	Used in
P-branch absorption model	$$\alpha(J) = C\tilde{\nu}(J)(2J+1)\exp\left[-\frac{hcBJ(J+1)}{k_B T}\right]$$	P-branch intensity modelling

Use of AI Assistance

ChatGPT was used as an auxiliary tool during the preparation of this report. It was used to support the overall structure of the report, assist with wording and clarity, help identify inconsistencies in the draft, and provide guidance for data analysis and graph plotting. All experimental data, numerical calculations, plots, final interpretations, and conclusions were checked and verified by the authors. ChatGPT was not used as a primary scientific source for the physical theory or reference values.